HTML Forms

Kassawmar M.
Web Programming

- Traditionally, the term 'form' has referred to a printed document that contains spaces for you to fill in information.
- HTML borrows the concept of a form to refer to different elements that allow you to collect information from visitors to your site.
- Provides two-way communication between web servers and browsers.
  - Demand for most of the emerging applications.
  - Providing dynamic contents.
- What is a HTML FORM?
  - A form basically contains boxes and buttons

Cont…

- Real-life examples
  - Search engines
  - On-line purchase of items
  - Registration
- The form allows a user to fill up the blank entries and send it back to the owner of the page.
  - Called SUBMITTING the form.

FORM Tags and Attributes

- Several tags are used in connection with forms:

  <form> …… </form>

  <input>

  <textarea> …… </textarea>

  <select> …… </select>

<FORM> …… </FORM> Tag

- This tag is used to bracket a HTML form.
  - Includes attributes which specify where and how to deliver filled-up information to the web server.
- Two main attributes:
  - METHOD
  - ACTION

Cont…

- METHOD:
  - Indicates how the information in the form will be sent to the web server when the form is submitted.
  - Two possible values:
    - GET: concatenates all field names and values in a single large string.
      - The parameters are visible in the URL itself.
      - Suitable for small amounts of data.
      - Limited length for the URL (usually around 2048 characters).
      - Not secure for sensitive data since the data is visible in the URL

Cont…

- POST: causes a form's contents to be parsed one element at a time.
  - Data is not visible in the URL.
  - Suitable for large amounts of data.
  - No limitation on the length of data.
  - More secure compared to GET because data is not visible in the URL.

Cont…

- ACTION:
  - Specifies the URL of a program on the origin server that will be receiving the form's inputs.
  - Traditionally called Common Gateway Interface (CGI).
  - The specified program is executed on the server, when the form is submitted.
    - Output sent back to the browser.

Cont…

- Typical usage:

```
<FORM METHOD="POST" ACTION="cgi-bin/myprog.pl">
……..
……..
</FORM>
```

<INPUT> Tag

- This tag defines a basic form element.
- Several attributes are possible:
  - TYPE
  - NAME
  - SIZE
  - MAXLENGTH
  - VALUE
  - SRC
  - ALIGN

Cont…

- **TYPE**:
  - Defines the kind of element that is to be displayed in the form.
    - "**TEXT**" – defines a text box, which provides a single line area for entering text.
    - "**RADIO**" – radio button, used when a choice must be made among several alternatives (clicking on one of the buttons turns off all others in the same group).
    - "**CHECKBOX**" – similar to the radio buttons, but each box here can be selected independently of the others.

Cont…

- "**PASSWORD**" – similar to text box, but characters are not shown as they are typed.
- "**HIDDEN**" – used for output only; cannot be modified (mainly used to refer to choices that have already been made earlier).
- "**IMAGE**" – used for active maps. When the user clicks on the image, the (x,y) co-ordinates are stored in variables, and are returned for further processing.
- "**SUBMIT**" – creates a box labeled Submit; if clicked, the form data are passed on to the designated CGI script.
- "**RESET**" – creates a box labeled Reset; if clicked, clears a form's contents.

Cont…

- NAME:
  - Specifies a name for the input field.
  - The input-handling program in reality receives a number of (name,value) pairs.
- SIZE:
  - Defines the number of characters that can be displayed in a TEXT box without scrolling.
- MAXLENGTH:
  - Defines the maximum number of characters a TEXT box can contain.

Cont…

- VALUE:
  - Used to submit a default value for a TEXT or HIDDEN field.
  - Can also be used for specifying the label of a button (renaming "Submit", for example).
- SRC:
  - Provides a pointer to an image file.
  - Used for clickable maps.
- ALIGN:
  - Used for aligning image types.
  - ALIGN = TOP | MIDDLE | BOTTOM

<TEXTAREA> … </TEXTAREA> Tag

- Can be used to accommodate multiple text lines in a box.
- Attributes are:
  - NAME: name of the field.
  - ROWS: number of lines of text that can fit into the box.
  - COLS: width of the text area on the screen.

<SELECT> …. </SELECT> Tag

- Used along with the tag <OPTION>.
- Used to define a selectable list of elements.
  - The list appears as a scrollable menu or a pop-up menu (depends on browser).
- Attributes are:
  - NAME: name of the field.
  - SIZE: specifies the number of option elements that will be displayed at a time on the menu. (If actual number exceeds SIZE, a scrollbar will appear).

Cont…

- MULTIPLE: specifies that multiple selections from the list can be made.

```
<FORM …….>
  ……..
  Languages known:
  <SELECT NAME="lang" SIZE=4>
    <OPTION> English
    <OPTION> Amharic
    <OPTION> Afan Oromo
    <OPTION> Tigrigna
  </SELECT>
</FORM>
```

Example 1

```
<HTML> <HEAD>
    <TITLE> Using HTML Forms </TITLE> </HEAD>
  <BODY TEXT="#FFFFFF"  BGCOLOR="#0000FF“ LINK="#FF9900"
    VLINK="#FF9900“ ALINK="#FF9900">
  <CENTER><H3> Student Registration Form </H3>
  </CENTER>
    Please fill up the following form about the courses you will register
      for this Semester.
<FORM METHOD="POST" ACTION="/cgi/feedback">
  <P> name: <INPUT name="name" TYPE="TEXT“ SIZE="30"
    MAXLENGTH="50">
  <P> roll number: <INPUT name="rollno“ TYPE="TEXT" SIZE="7">
  <P> course numbers:
      <INPUT name="course1" TYPE="TEXT" SIZE="6">
      <INPUT name="course2" TYPE="TEXT" SIZE="6">
      <INPUT name="course3" TYPE="TEXT" SIZE="6">
  <P> <P> press SUBMIT when done.
  <P> <input type="submit">
      <Input type="reset">
  </Form>  </body>  </html>
```

Grouping Form Elements

- **<fieldset>**
  - You can group related form controls together inside the **<fieldset>** element.
  - This is particularly helpful for longer forms.
  - Most browsers will show the **fieldset** with a line around the edge to show how they are related.
  - The appearance of these lines can be adjusted using CSS.
- **<legend>**
  - The **<legend>** element can come directly after the opening **<fieldset>** tag and contains a caption which helps identify the purpose of that group of form controls.

Cont…

```
<form method="post" action="registration.php">
    <fieldset>
     <legend><h1>well come to registration form</h1></legend></br>
          user fname:<input type="text" name="fname" /><br>
            user lname:<input type="text" name="lname" /><br>
             password:<input type="password" name="password" /><br>
         <input type="submit" value="submitt"  />
        </form>
```

html5: Form Validation

- You have probably seen forms on the web that give users messages if the form control has not been filed in correctly; this is known as form validation.
- Traditionally, form validation has been performed using JavaScript But HTML5 is introducing validation and leaving the work to the browser.
- Validation helps ensure the user enters information in a form that the server will be able to understand when the form is submitted.
- Validating the contents of the form before it is sent to the server the helps:

Cont..

- Reduce the amount of work the server has to do
- Enables users to see if there are problems with the form faster than if validation were performed on the server.
- It provides immediate feedback to users about the correctness of the data they've entered.
- An example of HTML5 form validation is the **required** attribute, which can be used on any form element that the user is expected to fill in.
- This HTML5 attribute does not need a value, but in HTML4 all attributes must have a value. So, some people give this attribute a value of **required**.
- By set the type pattern, maxlength, minlength. Like pattern=[A-Za-z];

HTML 5 DATE INPUT

- **<input>**
  Many forms need to gather information such as dates, email addresses, and URLs. This has traditionally been done using text inputs.
- HTML5 introduces new form controls to standardize the way that some information is gathered.
- **type="date"**
  If you are asking the user for a date, you can use an **<input>** element and give the **type** attribute a value of **date**.
- This will create a date input in browsers that support the new HMTL5 input types.

html5: Email & URL Input

- **<input>**
  HTML5 has also introduced inputs that allow visitors to enter **email addresses** and **URLs.**
- **type="email"**
  If you ask a user for an email address, you can use the email input. Browsers that support HTML5 validation will check that the user has provided information in the correct format of an email address.
- Some smart phones also optimize their keyboard to display the keys you are most likely to need when entering an email address (such as the @ symbol).

Cont…

- **Type ="url"**
  A URL input can be used when you are asking a user for a web page address.
- Browsers that support HTML5 validation will check that the user has provided information in the format of a URL.
- Some smart phones also optimize their keyboard to display the keys you are most likely to need when entering a URL.

HTML 5 FOR SEARCH INPUT

- **input type="search"**
  To create the HTML5 search box the **<input>** element should have a **type** attribute whose value is **search**.
- **placeholder**
  On any text input, you can also use an attribute called **placeholder** whose value is text that will be shown in the text box until the user clicks in that area.

How Forms Work

- A user fills in a form and then presses a button to submit the information to the server.
- The name of each form control is sent to the server along with the value the user enters or selects.
- The server processes the information using a programming language such as PHP, C#, VB.net, or Java. It may also store the information in a database.
- The server creates a new page to send back to the browser based on the information received.

HTML Form Controls

- There are several types of form controls that you can use to collect information from visitors to your site.
- **Adding text:**
  - **Text input (single-line)**
    Used for a single line of text such as email addresses and names using HTML **<input>** tag.
  - **Password input**
    Like a single line text box but it masks the characters entered. They are also created using HTMl **<input>** tag.
  - **Text area (multi-line)**
    For longer areas of text, such as messages and comments using HTML **<textarea>** tag

Cont…

- **Making Choices**
  - Radio buttons
    For use when a user must select one of a number of options.
  - Checkboxes
    When a user can select and unselect one or more options.
  - Drop-down boxes
    When a user must pick one of a number of options from a list.
- Submitting Forms:
  - **Submit buttons**

    To submit data from your form to another web page.

    The submit button is used to send a form to the server.

    It has name and value(attributes)
  - **Image buttons**
    Similar to submit buttons but they allow you to use an image.

Multiple-Line Text Input Controls

- Multi-line input controls are created using HTML **<textarea>** tag
- list of attributes for **<textarea>** tag.
  - **Name:** Used to give a name to the control
  - **rows**:the number of rows of text area box.
  - **Cols**:the number of columns of text area box

Checkbox Control

- Checkboxes are used when more than one option is required to be selected.
- They are also created using HTML **<input>** tag but type attribute is set to **checkbox.**

Cont..

- list of attributes for **<checkbox>** tag.
  - **Type**:Indicates the type of input control
  - **Name:**Used to give a name to the control which is sent to the server to be recognized and get the value.
  - **value**: The value that will be used if the checkbox is selected.
  - **Checked:** Set to checked if you want to select it by default.

Radio Button Control

- Radio buttons are used when out of many options, just one option is required to be selected.
- They are also created using HTML **<input>** tag but **type** attribute is set to **radio**.
- Attributes are similar with **checkbox** except the value of **type** attribute
- The **name** of radio button should be the same

Select Box Control

- Also called ***drop down*** box which provides option to list down various options in the form of drop down list, from where a user can select one or more options.
- Attributes

  attributes of <select> tag:

  - **Name:** Used to give a name to the control
  - **Size**: used to present a scrolling list box.
  - **Multiple:**If set to "multiple" then allows a user to select multiple items from the menu.

Cont…

- Attributes of **<option>** tag
  - **value** :The value that will be used if an option in the select box is selected.
  - **Selected:** Specifies that this option should be the initially selected value when the page loads.
  - **Label:**An alternative way of labeling options

File Upload Box

- allow a user to upload a file to your web site
- Example

```
<form action="“ method="post">

<p>Upload your song in MP3 format: </p>
<input type="file" name="user-song" /><br />
<input type="submit" value="Upload" />

</form>
```

Button Controls

- You can create a clickable button using **<input>** tag by setting its type attribute to **button**. The type attribute can take the following values:
- **Submit:** creates a button that automatically submits a form.
- **reset** : creates a button that automatically resets form controls to their initial values.
- **Button:** This creates a button that is used to trigger a client-side script when the user clicks that button.
- **Image:**This creates a clickable button but we can use an image as background of the button.

Hidden Form Controls

- used to hide data inside the page
- hides inside the code and does not appear on the actual page.
- Can creates by ***<input>*** with type attribute value is **hidden**

HTML5: adding Video To Your Pages

<video>

- The <video> element has a number of attributes which allow you to control video playback:
  - Src: this attribute specifies the path to the video.
  - Poster: this attribute allows you to specify an image to show while the video is downloading or until the user tells the video to play.
  - width, height: these attributes specify the size of the player in pixels.
  - Controls When used, this attribute indicates that the browser should supply its own controls for playback.

Cont..

- Autoplay When used, this attribute specifies that the file should play automatically.
- Loop When used, this attribute indicates that the video should start playing again once it has ended.
- **Preload** This attribute tells the browser what to do when the page loads. It can have one of three values:
  - ✓ **none** The browser should not load the video until the user presses play.
  - ✓ **Auto** The browser should download the video when the page loads.

Cont..

- <!DOCTYPE html>

  <html>

  <head>

  <title>Adding HTML5 Video</title>

  </head> <body>

  <video src="video/puppy.mp4“ poster="images/puppy.jpg“

  width="400" height="300“ preload controls loop>

  A video of a puppy playing in the snow

  </video> </body> </html>

HTML5: multiple Video sources

- **<source>** To specify the location of the fie to be played, you can use the **<source>** element inside the **<video>** element. (This should replace the **src** attribute on the opening **<video>** tag.) You can also use multiple **<source>** elements to specify that the video is available in different formats.

  Src This attribute specifies the path to the video.

  Type You should use this attribute to tell the browser what format the video is. Otherwise, it will download some of the video to see if it can play the fie (which will take time and bandwidth).

example

- <!DOCTYPE html>
  <html> <head>
  <title>Multiple Video Sources</title>
  </head> <body>
  <video poster="images/puppy.jpg" width="400“ height="320" preload controls loop>
  <source src="video/puppy.mp4" type='video/mp4;codecs="avc1.42E01E, mp4a.40.2"' />
  <source src="video/puppy.webm" type='video/webm;codecs="vp8, vorbis"' />
  A video of a puppy playing in the snow
  </video> </body> </html>

Adding audio To web Pages

- By far the most popular format for putting audio on web pages is MP3.
- **<audio>** HTML5 introduced the **<audio>** element to include audio files in your pages. As with HTML5 video, browsers expect different formats for the audio. The **<audio>** element has a number of attributes which allow you to control audio playback:
- **src** This attribute specifies the path to the audio fie.
- **Controls** This attribute indicates whether the player should display controls. If you do not use this attribute, no controls will be shown by default. You can also specify your own controls using JavaScript.

Cont…

- **Autoplay** The presence of this attribute indicates that the audio should start playing automatically. (It is considered better practice to let visitors choose to play audio.)
- **preload**

  This attribute indicates what the browser should do if the player is not set to **autoplay**.
- **loop**

  This attribute specifies that the audio track should play again once it has finished.

Example

```
<!DOCTYPE html>
<html> <head>
<title>Adding HTML5 Audio</title>
</head> <body>
<audio src="audio/test-audio.ogg" controls autoplay>
<p>This browser does not support our audio format. </p>
</audio>
</body>
</html>
```

HTML5: multiple audio sources

- **<source>**

  It is possible to specify more than one audio fie using the **<source>** element between the opening **<audio>** and closing **</audio>** tags (instead of the **src** attribute on the opening **<audio>** tag).

- This is important because different browsers support different formats for audio files.

- You could also provide a Flash alternative for older browsers that do not support the **<audio>** element.

Cont…

- **src**

  The **<source>** element uses the **src** attribute to indicate where the audio fie is located.

  Type At the time of writing, the **type** attribute was not commonly being used on the **<source>** element in the same way it was for the **<video>** element.

Example

- <!DOCTYPE html>

  <html> <head>

  <title>Multiple Audio Sources</title>

  </head> <body>

  <audio controls autoplay>

  <source src="audio/test-audio.ogg" />

  <source src="audio/test-audio.mp3" />

  This browser does not support our audio format.

  </audio> </body> </html>

Block Elements

- Block elements appear on the screen as if they have a line **break** before and after them.
- For example

  <p>,

  <h1>, <h2>, <h3>, <h4>, <h5>, <h6>,

  <ul>, <ol>, <dl>,

  <pre>, <hr />, <blockquote>, and <address>

They all start on their own new line, and anything that follows them appears on its own new line.

Grouping text & Elements in a block

** **

- The ** ** element allows you to group a set of elements together in one block-level box.
- For example, you might create a ** ** element to contain all of the elements for the header of your site (the logo and the navigation), or you might create a ** ** element to contain comments from visitors.
- In a browser, the contents of the ** ** element will start on a new line.

Cont…

- Using an **id** or **class** attribute on the ** ** element, however, means that you can create CSS style rules to indicate how much space the ** ** element should occupy on the screen and change the appearance of all the elements contained within it.
- It can also make it easier to follow your code if you have used ** ** elements to hold each section of the page.

Cont…

- Example

```
<div id ="header">
<img src="images/logo.gif" alt="Anish Kapoor" />
<ul> <li><a href="index.html">Home</a></li>
<li><a href="biography.html">Biography</a></li>
<li><a href="works.html">Works</a></li>
<li><a href="contact.html">Contact</a></li>
</ul>  </div><!-- end of header -->
```

Inline Elements

- Inline elements, on the other hand, can appear within sentences and do not have to appear on a new line of their own.
- The **<b>, <i>, <u>, , , <sup>, <sub>, <big>, <small>, <li>, <ins>, <del>, <code>, <cite>,**elements are all inline elements.

Grouping text & Elements inline

- ****
  The **** element acts like an inline equivalent of the ** ** element.
- It is used to either:
  - Contain a section of text where there is no other suitable element to differentiate it from its surrounding text
  - Contain a number of inline elements
- The most common reason why people use **** elements is so that they can control the appearance of the content of these elements using CSS.
- You will usually see that a **class** or **id** attribute is used with **** elements:
  - To explain the purpose of this **** element
  - So that CSS styles can be applied to elements that have specific values for these attributes

Cont…

id attribute

- Every HTML element can carry the **id** attribute. It is used to uniquely identify that element from other elements on the page.
- Its value should start with a letter or an underscore (not a number or any other character).
- It is important that no two elements on the same page have the same value for their **id** attributes (otherwise the value is no longer unique).
- The **id** attribute is known as a global attribute because it can be used on any element.(we will see more on CSS section)

Class attribute

- Every HTML element can also carry a **class** attribute.
- Sometimes, rather than uniquely identifying one element within a document, you will want a way to identify several elements as being different from the other elements on the page.
- To do this you can use the **class** attribute.
- Its value should describe the class it belongs to.
- The **class** attribute on any element can share the same value.

End of Chapter Two

Thanks for you attention!!!